

ncurses libraries, so we can also use the mouse in console applications. There also exists a stripped-down lightweight version of Dialog known as *lxdialog*, which is used in Linux kernel configuration.

Xdialog is the functional counterpart of the Dialog program that produces various widgets in X. It uses the GTK+ library for its functionality, which means that we can smoothly integrate Xdialog in desktop environments like GNOME. So, using Xdialog we can incorporate graphics mode widgets in our command-line applications.

Also, for KDE, there exists a similar program like Kdialog known as *kdialog*, but we'll leave that out of this article. In fact, after getting the feel of Dialog and Xdialog, you can handle similar utilities pretty much in the same manner. So readers are encouraged to explore *kdialog* from the tutorial link provided under the *References* section at the end of the article.

Installation

To my surprise, both Dialog and Xdialog came preinstalled in Puppy Linux 4.2.1. On Ubuntu 9.10, the easiest way is to issue the following command on a text console:

```
sudo apt-get install dialog
```

Ubuntu 9.10 also contains a utility known as 'Whiptail', which provides some of the functionality of Dialog, but is based on the Newt library. I found Whiptail very limited with respect to the widgets it offered and the configuration options it provided. We will discuss it later on in the article.

Installation of Xdialog on Ubuntu 9.10 involved more work. I had to install it from source as there wasn't any installation candidate for it in the software repository. To install Xdialog from source, I first had to download and install the following *.deb* packages in the same order as shown below (note that Xdialog requires the *gtk-1.2* series only and not other versions of GTK):

1. *libgtk1.2-common_1.2.10-18_all.deb*
2. *libgl1.2-1.2.10-10.1build1_amd64.deb*
3. *libgtk1.2-1.2.10-18_amd64.deb*
4. *libgl1.2-dev_1.2.10-10.1build1_amd64.deb*
5. *libgtk1.2-dev_1.2.10-18_amd64.deb*

Once done, I downloaded the latest Xdialog sources from its home page, uncompressed its tar archive and moved to the source directory that had been created by typing the following commands:

```
tar -xvf Xdialog-2.3.1.tar.bz2
cd Xdialog-2.3.1
```

To finally build and install Xdialog, I ran the following command:

```
./configure && make && sudo make install
```

So we are now ready to play around with Xdialog.



Figure 1: infobox widget in action

First, let's learn Dialog

Getting started with Dialog is straightforward—think of it as a command provided by our shell. We provide different common widget options with it to do different widgets operations. The general form of *dialog* usage is as follows:

```
dialog <common options> widget <widget options>
```

The *<common options>* are optional settings that modify the appearance and behaviour of Dialog widgets. The *<widget options>* are combinations of the required and optional settings to create widgets. Note that in the man pages, the widgets options are mentioned as box options.

There are many options we can use with Dialog, but here we will experiment with just a few. I encourage you to tweak the remaining options.

Some common options that we will use along with their functions are:

- *–backtitle titlestring* – The background canvas of Dialog widgets usually doesn't have any text. By using this option, we can show some text on it.
- *–title titlestring* – By using this option we can add some text as a title to widgets.
- *–stdout* – By default, some of the widgets that emit an output, write it to the screen. We can use this option if we want to gather the output, without having it written to the screen.
- *–begin ypos xpos* – By default, all the widgets are positioned towards the centre of the text console screen. We can change that using this option, by stating the *y* and *x* positions from the top left corner of the text console.
- *–timeout sec* – All widgets, by default, only return when Esc is pressed, if any of the widget buttons are selected and Enter is pressed, or if the mouse is clicked over any of the buttons. Using this option, we can tweak the default behaviour by specifying the interval in seconds, after which the widget automatically returns.
- *–sleep sec* – This one inserts a delay for the specified number of seconds after processing the Dialog widget.
- *–no-shadow* – By default, a shadow is shown to the right and bottom of the widgets to create an illusion of depth. Using this option, we can turn it off.